2026 Basic Abstract Algebra Study

LECTURE 1 HOMEWORKS

SOLUTION - Q5

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Read this carefully before moving on.

Please, clearly specify whether the statement in question is *true* or *false*, for the *prove or disprove* type of questions. As always, it is highly encouraged for you to write each step clearly with full sentences.

As a demonstration of a proof answer, consider the following example.

The above statement is **true**.

Proof. ... □

As a demonstration of a disproof answer, consider the following example.

The above statement is **false**.

Proof. ... □

Note 1 (Notations)

The following notations will be used throughout the following questions.

- $\mathbb{Z}$: A set of all integers.
- $\mathbb{Z}^+$: A set of all positive integers. i.e., $\mathbb{Z}^+ = \{1, 2, 3, ...\}$.
- $\mathbb{Z}_n$: A set of non-negative integers from 0 to $n - 1$ for some positive integer n. For instance, $\mathbb{Z}_4 = \{0, 1, 2, 3\}$.
- $n\mathbb{Z}$: A set given as $\{nx \mid x \in \mathbb{Z}\}$ for an integer n. i.e., the set of all multiples of n in $\mathbb{Z}$. For instance, $5\mathbb{Z} = \{..., -10, -5, 0, 5, 10, ...\}$.
- $\mathbb{Q}$: A set of all rational numbers.
- $\mathbb{R}$: A set of all real numbers.
- $\mathbb{R}^+$: A set of all positive real numbers.
- $\mathbb{C}$: A set of all complex numbers.
- $A + B$: A set given as $\{a + b \mid a \in A, b \in B\}$. For instance, $\mathbb{Z}_2 + \mathbb{Z}_3 = \{(0 + 0), (0 + 1), (0 + 2), (1 + 0), (1 + 1), (1 + 2)\} = \{0, 1, 2, 3\}$

Definition 2 (Subset)

Given two sets A and B, A is said to be a subset of B if the following condition holds.

If x is an element of A, then x is an element of B.

Definition 3 (Extensionality)

Two sets A and B are said to be equal if and only if A is a subset of B and B is a subset of A. This is called the *axiom of extensionality*. If two sets A and B are equal, we write $A = B$.

Q1.

EASY

Let A and B be two sets. Let $A(x)$ mean x is an element of A, and $B(x)$ mean x is an element of B.

Which of the following is equivalent to saying that $A \subset B$?

◯ **1.** $B(x) \to A(x)$ ◯ **2.** not $A(x) \to B(x)$ ◯ **3.** not $A(x) \to$ not $B(x)$
◯ **4.** $A(x) \to$ not $B(x)$ ◯ **5.** not $B(x) \to$ not $A(x)$

Definition 4 (Roots of unity)

Let μ_n denote the set of all complex solutions to an equation of form $x^n = 1$ for a positive integer n. Elements of such a set are called *n-th roots of unity.*

For instance, $\mu_2 = \{1, -1\}$.

Q2.

EASY

Show that $\mu_2 \subset \mu_4$.

Q3.

EASY

Prove or disprove that $\mathbb{Z} \subset n\mathbb{Z} + m\mathbb{Z}$ for any two distinct positive integers n and m.

Hint 5. *Provide concrete n and m that make up a counterexample. Try justifying your answer by using the definition of set builder notation.*

Q4.

HARD

Prove or disprove that $2\mathbb{Z} + 3\mathbb{Z} = \mathbb{Z}$.

Hint 6. *cf. lecture note.*

Q5.

HARD

Show using Definition 2 that for any two positive integers n and m, $\mu_n \subset \mu_{nm}$.

Hint 7. *Of course, it is a generalization of **Q2** .*

Recall the Definition 2 that $\mu_n \subset \mu_{nm}$ essentially means the following.

If x is a solution to $x^n = 1$, then it is also a solution to $x^{nm} = 1$.

Proof. For two positive integers n and m, assume x is an element of μ_n. By the definition, we have $x^n = 1$. By raising both sides to the m-th power, we get $(x^n)^m = x^{nm} = 1^m = 1$. From here, we conclude that x is a solution to the equation $x^{nm} = 1$, thus $x \in \mu_{nm}$.

Therefore, μ_n is a subset of μ_{nm}. □

HARD

Q6.

Show by Definition 3 that $\mathbb{Z}_n + \mathbb{Z}_n = \mathbb{Z}_{2n-1}$ for any positive integer n.

Hint 8. *cf. PCSA Q4.*

Q7.

HARD

Prove or disprove that if $\mathbb{Z} = n\mathbb{Z}$ for an integer n, then either $n = 1$ or $n = -1$.

Hint 9. *What are the factors of* 1 *and* -1*?*

Definition 10 (Symmetric difference)

$A \triangle B$ is a binary set operation given by $(A \setminus B) \cup (B \setminus A)$ and is called a *symmetric difference* between A and B.

Q8.

LEVI

Problem by @escentia07.

Prove or disprove that $A \triangle B \subset C$ if and only if $A \cup B \subset (A \cap B) \cup C$.

Q9.

LEVI

For an integer n, let $M(k)$ denote the set $\{x \mid x \equiv k \pmod{n}\}$. Let P be the set given by $P = \{M(k) \mid k \in \mathbb{Z}\}$.

(a) Show that every two distinct elements of P are disjoint.

Hint 11. *Use Definition 3 and contraposition.*

(b) Show that the following holds for any two integers a and b by Definition 3.

$$M(a) + M(b) = M(a + b)$$

Q10.

LEVI

Problem by @escentia07.

Let $f(x) = x^{2121} + x - 1$

How many real solutions does the equation $f^{2026}(x) = x$ have? Justify your answer.

Answer: __________

Q11.

LEVI

Let T be the set $\{z \in \mathbb{C} \mid |z| = 1\}$ and μ_∞ be the set of all roots of unity as follows.

$$\mu_\infty = \bigcup_{n=1}^{\infty} \mu_n = \mu_1 \cup \mu_2 \cup \dots$$

Prove or disprove that $T = \mu_\infty$.

End.

You may now submit your work.

Optional evaluation sheet

Unlike PCSA, submitting the sheet is **optional** for homeworks.

For each question so far, you can evaluate it based on the following criteria:

- If you failed to understand the question, mark × in the Evaluation column.
- If you understood the question, but didn't know how to solve it, mark ? in the Evaluation column.
- If you understood the question and are sure you have solved it, mark ○ in the Evaluation column.

No.	Evaluation	No.	Evaluation
Q1		Q7	
Q2		Q8	
Q3		Q9	
Q4		Q10	
Q5		Q11	
Q6			

There are these additional questions as well; feel free to answer them.

(a) Which was the hardest for you, among those you have solved? __________

(b) Which seemed to be the hardest for you, among those you haven't solved? __________

(c) Which was your favorite question, if you had to choose one? __________